

Efficient RNN Models for IOT Intrusion Detection System

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Abstract— Due to the growing number of network devices, traffics and services, designing robust Intrusion Detection System (IDS) become a crucial need in the face of complex and various network attacks as a protective measure from hackers and cybercriminals. However, the traditional Machine Learning (ML) approach shows success in many research topics but with the increase in the amount of data and the diversification of network threats methods, it seems to lack reliability and accuracy. Therefore, considering a large amount of real-world cyber traffic, Deep Learning (DL) may be able to extract big data features more effectively. In this paper, we suggest an intrusion detection system for the Internet of Things (IoT) network based on Deep Learning to recognize various assault types for both binary and multiclass classification using two variants of Recurrent Neural Network (RNN) models long short-term memory (LSTM) and Bi-directional LSTM (BiLSTM). We have experimented the models with CSE-CIC-IDS2018, which is the newest comprehensive network traffic dataset. Accuracy, precision, recall, and F1 score are a few performance criteria where the suggested approach clearly excels. After comparison, we can infer that Bi-directional LSTM outperforms LSTM and other existing efforts in the literature. The accuracy of the experimental results was high, coming in at 98.62%.

Keywords— IDS, DL, RNN, LSTM, BiLSTM, Network Traffic

I. INTRODUCTION

In recent years, with the widespread use of the Internet in every aspect of people's daily life and work and the growth of intelligent devices, the Internet of Things (IoT) is an exciting method for connecting and accessing any device via the internet. It has been stormed to various fields such as industry, smart homes, business, healthcare and medicine, agriculture, energy management, and the military [1]. Due to the volume and variety of breaches on company networks and systems, cyber security has grown to be a serious issue. IDS has become crucial in such situations to reduce and avoid the impact of cybercrimes and assist in defending systems from various types of harmful cyberattacks [2].

Traditional Machine Learning (ML) [3] based intrusion detection algorithms including Random Forests (RF), AdaBoost, Support Vector Machines (SVM), K-Means, and K-Nearest Neighbors (KNN) have been widely used in developing

IDS [4]. However, even though the enormous efforts of researchers, traditional machine learning [5] algorithms are getting more and more unfit for use in network application contexts because of the raise of datasets regarding size and type and challenges to be addressed in performance metrics such as improved detection rate, less false positives, and the capacity to identify unidentified threats.

Recently, researchers have successfully used Deep Learning (DL) based techniques in a range of applications, including handwritten character identification, emotion detection, picture recognition, and computer vision. Also, some studies have used the DL algorithm on IDS, such as Convolutional Neural Networks (CNN), Deep Belief Networks (DBN), Deep Neural Networks (DNN), and Recurrent Neural Networks (RNN) [6]. In terms of accuracy and capacity to handle enormous volumes of data, the DL approach or its combination has proven considerably superior to ML models [7], [8].

This work focuses on the use of LSTM and BiLSTM two variants of RNN and their application to an IDS model. RNN is advantageous for handling time series tasks and is capable of addressing problems with long-term reliance. However, the gradient vanishing problem is RNN's most difficult task. To deal with this, researchers have put up a number of solutions, the most prominent of which are the LSTM network and its variant BiLSTM. Also, the Gated Recurrent Unit (GRU) network is one of the commonly used ones [9].

The IDS proposed in this research uses LSTM and BiLSTM models to identify various attack types for binary and multiclass classification using the CSE-CICIDS2018 datasets, the developed system is trained and tested, and its accuracy is assessed.

The rest of this document is structured as follows. Part II compiles the results of the previous literature reviews. Part III describes the suggested LSTM and BiLSTM models, along with a brief discussion of the dataset and model architecture. The findings of our study are given and discussed in Section IV. This essay's conclusion, Section V, addresses potential study avenues.

II. RELATED WORK

In regard to Network Intrusion Detection issues, DL has long been popular. For instance, authors in [10] compared the effectiveness of three distinct DL models: LSTM, GRU, and

RNN, while handling the issue of anomaly detection using CICIDS2017 and CSE-CICIDS2018 datasets.

In [11], The study constructed NIDS models based on stacked LSTMs and Bi-Directional LSTMs using the UNSW NB15 and BoT-IoT datasets. The accuracy of the Bi-Directional LSTM model was 96.41%, while the accuracy of the Stacked LSTM model was 96.60%. Similarly, the accuracy of the two models on the BoT-IoT dataset was 99.99%.

In [12], B. B. Borisenko et al. Examined network traffic multiclass classification tasks that were performed using multilayer perceptron and LSTM on the CSE-CICIDS2018 dataset. The evaluation focused on the effectiveness of individual computer attack detection given that LSTM is able to identify many attacks that the multilayer perceptron was unable to identify.

In order to evaluate the effectiveness of the BiLSTM and LSTM models, a BiLSTM-based IDS using the NSL-KDD dataset was evaluated by Imrana et al. in [13]. The LSTM model generates a higher false alarm rate as compared to the BiLSTM model

In [14], Alkahtani et al. designed a deep CNN-LSTM using N-BaIoT dataset for identifying botnet breaches on nine different IoT devices. With an average accuracy of 90%, their technique excels in recognizing most attacks from diverse devices.

In [15], Using a BiLSTM and CNN hybrid model, the authors present an IDS for detecting abnormalities in a smart home network. The CNN correctly extracts the data characteristics, and the BiLSTM maintains the learned knowledge over time. Any network gateway of a smart home can use the proposed model. The suggested BiLSTM-CNN hybrid model performs better than the most advanced IoT device anomaly detection methods.

In [16], the paper proposed a CNN-BiLSTM model using the NSL-KDD data set. The experimental findings demonstrate that the performance is superior to previous results in the literature in terms of accuracy (over 96%), and detection rate (97.1%).

III. METHODOLOGY

We begin by delving into the LSTM structure. Next, we go into great depth on the BiLSTM structure. The CSE-CICIDS2018 dataset, which was utilized to train our model, is further described.

A. Long short-term memory (LSTM)

LSTM is an RNN variety. It is suggested by Hochreiter et al. [17] to resolve the gradient vanishing problem. LSTM shows outstanding performance on a large variety of applications such as images and video, machine translation, and language modeling. A typical LSTM network architecture includes a memory cell and three regulators to govern data flow inside the LSTM: an input gate, an output gate, and a forget gate. Moreover, the Gated Recurrent Unit has controlled the LSTM unit (GRU) [18]. Fig. 1 describe the LSTM cell's architecture. The following steps must be taken in order to achieve an updated neural unit activation:

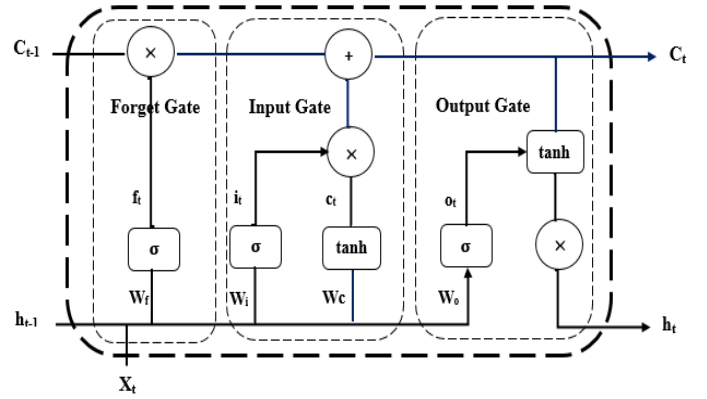


Fig. 1. Structure of LSTM Cell

$$i_t = \sigma(W_{xi}x_t + W_{hi}h_{t-1} + W_{ci}c_{t-1} + b_i) \quad (1)$$

$$f_t = \sigma(W_{xf}x_t + W_{hf}h_{t-1} + W_{cf}c_{t-1} + b_f) \quad (2)$$

$$c_t = f_t * c_{t-1} + i_t * \tanh(W_{xc}x_t + W_{hc}h_{t-1} + b_c) \quad (3)$$

$$o_t = \sigma(W_{xo}x_t + W_{ho}h_{t-1} + W_{co}c_{t-1} + b_o) \quad (4)$$

$$h_t = o_t * \tanh(c_t) \quad (5)$$

σ and $\tanh$ symbolize the activation functions, c is the cell state, i , f and o are the input, forget, and output gate, respectively. W represents the weight matrix while b defines the bias vector.

B. Bidirectional LSTM(BiLSTM)

BiLSTM[19] is an LSTM extension that improves model performance on challenges involving sequence categorization by learning in two directions at the same time. BiLSTM is advantageous since it trains two LSTMs instead of one when all timesteps in the input sequence are available. In determining the output, BiLSTM takes both forward and backward activation into account. The backward-working LSTM unit extracts the backward features of the input layer's data sequence, and the forward-working LSTM unit extracts the forward features of the input layer's data sequence. Data produced from the forward and backward transmission layers are combined in the output layer [20]. In Fig. 2, the BiLSTM structure is displayed.

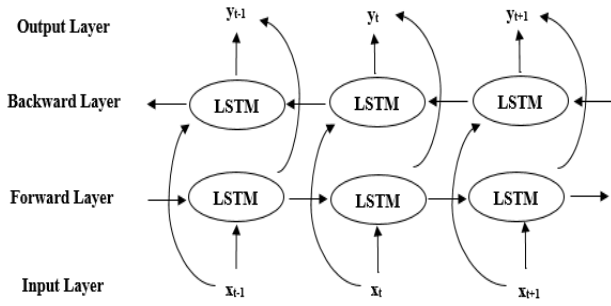


Fig. 2. BiLSTM structure

C. Dataset

1) *Dataset Description*: For this research, among the most current publicly accessible intrusion detection datasets was employed. The CSE-CIC-IDS2018 [21] is more intricate and reflective of real-world contemporary network traffic and current attack types. For this dataset, the Canadian Institute for Cybersecurity (CIC) and the Communications Security Establishment (CSE) worked together. The dataset includes seven various assault scenarios. It is made up of unprocessed PCAP network recordings and CICFlowMeter-V3 processed CSV files that give 80 features. Network traffics are recorded from Amazon Web Services (AWS) for ten days, at which time numerous attack scenarios and realistic traffic were assessed. The dataset, therefore, includes instances of both benign network traffic and typical network attacks [22].

2) *Dataset Preprocessing*: The dataset CSE-CIC-IDS2018 comprises 16 232 942 rows, each of which consists of 10 files, with 80 functions per row. For this experiment, only 3 files were used. They contain 3 145 725 data samples and 80 features, as shown in Table I. Moreover, 8 features including Timestamp, Protocol, PSH Flag Cnt, Init Fwd Win Byts, Flow Byts/s, Flow Pkts/s, and Label needed to be removed because they do not reflect any impact on network attack. Cleaning the data of null values and infinite values come next. It is allowed to use the infinity value as the column's highest value with some variance and a negligible impact on the effectiveness of model evaluation. To prevent such numbers from impacting the mode during training, it is imperative to change the NaN values that were eliminated from the samples [23]. Moreover, the values were adjusted to lie between 0 and 1. The labels are encoded using the one-hot encoding technique. 3 135 969 data samples total will be used for testing and training. After that, the dataset will be split into two subsets: training set (80%) and testing set (20%). The fact that this study uses both binary and multiclass classification is equally significant. As a result, we put the Benign label to 0 and the Attack labels to 1 (see Table II). We divide the label into six groups in order to do multiclass classification (Benign, Bot, FTP-BruteForce, SSH-BruteForce, DoS-GoldenEye, and DoS-Slowloris). The number of traffic records in our dataset for each type is shown in Tables II and III.

TABLE I. SUMMARY OF THE CSE-CIC-IDS2018 DATASET

Data File	Traffic Type	Number of Samples	Some of features
02-14-2018.csv	Benign	663,808	Dst Port, Protocol, Flow Duration, Tot Fwd Pts, Tot Bwd Pkts, TotLen Fwd Pkts, TotLen Bwd Pkts, Fwd Pkt Len Max,
	FTP-BruteForce	193,354	
	SSH-BruteForce	187,589	
02-15-2018.csv	Benign	988,050	
	DoS-GoldenEye	41,508	
	DoS-Slowloris	1,099	
03-02-2018.csv	Benign	758,334	
	BotAttack	286,191	
Total		3145725	

TABLE II. DETAILS OF CLASS LABELS ON BINARY CLASSIFICATION

Category	Label	Number of samples
Benign	0	2416337
Attack	1	719632

TABLE III. DETAILS OF CLASS LABELS ON MULTI CLASSIFICATION

Category	Label	Number of samples
Benign	0	2426087
Bot	1	286191
FTP-BruteForce	2	193360
SSH-BruteForce	3	187589
DoS GoldenEye	4	41508
DoS Slowloris	5	10990

D. Model Architecture

Following preprocessing, the reshaped CSE-CICIDS2018 datasets are sent to the LSTM and BiLSTM layers. The two models were built using the same construction, as illustrated in Fig. 3. An input layer followed by two recurrent neural network layers using the rectified linear unit (ReLU) and a dense layer using softmax as an activation function make up the model. The models' final output makes a prediction about what attack class the traffic pattern will belong to. In order to prevent our model from over-fitting the data, we lastly add a dropout probability of 0.3. After many essays, we are guided to choose various suitable parameters in the training phase [24] (see Table IV).

TABLE IV. TRAINING PARAMETERS VALUES

Parameters	Values
Epochs	10
Batch size	64
Learning rate	0.001
Optimizer	Adam(Adaptive Moment)
Cost function	Binary cross-entropy categorical cross-entropy

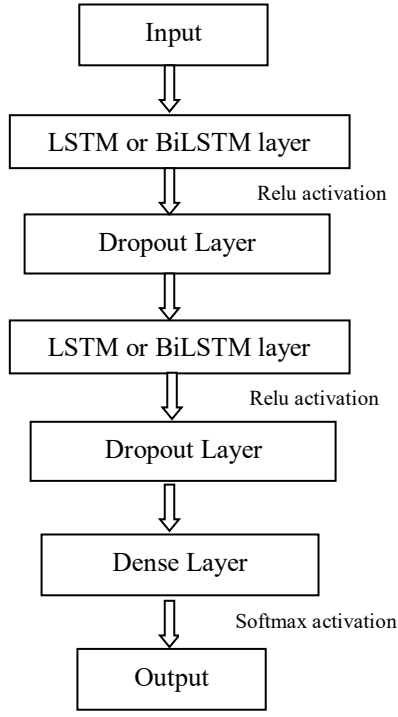


Fig. 3. LSTM and BiLSTM architecture models

IV. RESULTS AND DISCUSSIONS

A. Experimental Setup

All experiments are performed on windows 10 64-bit operating system environment using Python language (version. 3.7.13) with TensorFlow (version. 2.9.1), Keras(version. 2.9.0) framework, and various other library packages like pandas, numpy, matplotlib, and Sklearn. Moreover, running on a CPU with Intel(R) Core(TM) i7-8700U CPU 3.20 GHz, 16 GB memory with an Nvidia GeForce GT710 GPU.

B. Performance metrics

Various metrics are generated to evaluate IDS performance based on the following parameters:

- True Positive (TP): Number of correctly classified attack samples.
- False Positive (FP): Number of misclassified attack samples.
- True Negative (TN): Number of correctly classified benign samples.
- False Negative (FN): Number of misclassified benign samples.

The proposed LSTM and BiLSTM models were validated using accuracy, recall, precision, and F1 score. The formulas are presented in Equations 6, 7, 8, and 9.

$$Accuracy = \frac{TP + TN}{TP + FP + TN + FN} \quad (6)$$

$$Precision = \frac{TP}{TP + FP} \quad (7)$$

$$Recall = \frac{TP}{TP + FN} \quad (8)$$

$$F1Score = 2 \times \frac{Precision \times Recall}{Precision + Recall} \quad (9)$$

C. Binary Classification

In Binary Classification the model predicts whether the sample is an attack or belongs to a benign class. The effectiveness of the LSTM model is tested in this experiment using the confusion matrix illustrated in Fig. 4. (a). In comparison, the performance of the BiLSTM model is assessed using Fig. 4(b). Also, Figures 5 and 6 demonstrate the variations in accuracy and loss value of the training and validation sets of LSTM and BiLSTM for binary classification. The binary LSTM and binary BiLSTM both attained an accuracy of 98.30% and 98.62%, respectively. Besides, our model has good precision, recall, and F1 score with a few misclassified instances. The evaluation of binary classification using the LSTM and BiLSTM model's findings are summarized in Table V. As described, the BiLSTM model obtained a better precision for benign class, it's around 99% against 98% for LSTM and share the same performance of 98% for Attack class. Additionally, F1-Score is higher for BiLSTM by 0.01% than LSTM for the Attack class. Furthermore, our two models achieve a greater recall on the detection of benign labels, 100% and 99%, respectively, on LSTM and BiLSTM, however, the BiLSTM model improves the detection rate (recall) of the LSTM model by 0.02% on the Attack class.

TABLE V. LSTM AND BiLSTM MODELS PERFORMANCE FOR BINARY CLASSIFICATION

Model	LSTM			BiLSTM			
	Class	Precision	Recall	F1score	Precision	Recall	F1score
	Benign	0.98	1.00	0.99	0.99	0.99	0.99
	Attack	0.98	0.94	0.96	0.98	0.96	0.97

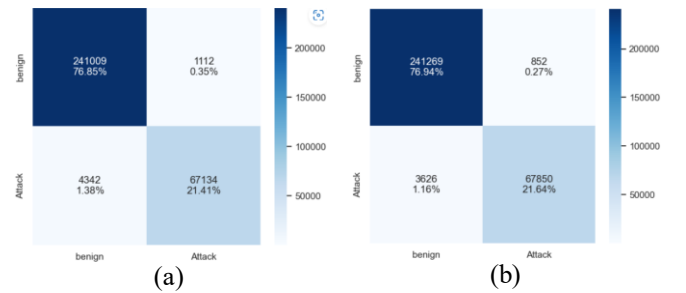


Fig. 4. (a) LSTM. (b) BiLSTM confusion matrix for binary classification

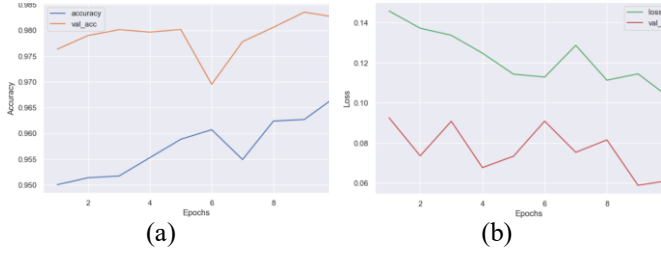


Fig. 5. (a) Accuracy. (b) Loss of binary classification using LSTM

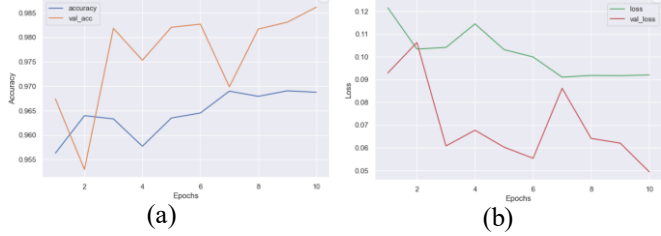


Fig. 6. (a) Accuracy. (b) Loss of binary classification using BiLSTM

D. Multiclass Classification

In multiclass classification, the dataset was classified as benign network traffic or one of the six attacks outlined in Table III. Figures 7 and 8 show the performance of the LSTM and BiLSTM models' multiclass classification models in terms of accuracy and loss during training and validation. Fig. 9 (a) shows the confusion matrix used to evaluate the LSTM. In contrast, Fig. 9 (b) depicts the matrix used to evaluate the BiLSTM. The accuracy of the CSE-CIC-IDS2018 dataset was measured at 95.01% for LSTM and 95.62% for BiLSTM models. Moreover, BiLSTM achieves the highest precision and F1-Score compared to LSTM. Also, both of the two models successfully detected the Benign, Bot, and DoS-GoldenEye labels. However, the predictions miss FTP-BruteForce, SSH-BruteForce, and DoS-Slowloris with a little preference for BiLSTM as shown in table VI.

TABLE VI. LSTM AND BiLSTM PERFORMANCE FOR MULTICLASS CLASSIFICATION

Model	LSTM			BiLSTM		
Class	Precision	Recall	F1score	Precision	Recall	F1score
Benign	0.98	1.00	0.99	0.99	0.99	0.99
Bot	1.00	0.99	1.00	1.00	0.99	1.00
FTP-BruteForce	0.79	0.20	0.40	0.89	0.31	0.46
SSH-BruteForce	0.63	0.21	0.31	0.72	0.44	0.55
DoS-GoldenEye	0.67	1.00	0.80	0.67	1.00	0.80
DoS-Slowloris	0.96	0.49	0.65	0.95	0.50	0.65

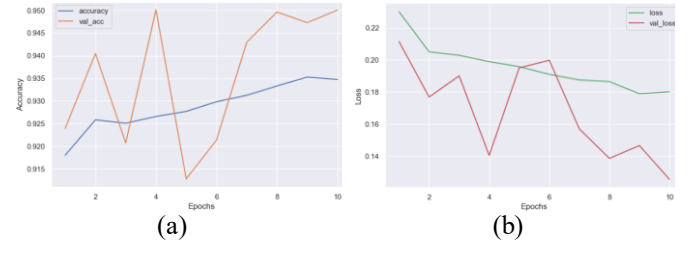


Fig. 7. (a) Accuracy. (b) Loss of multiclass classification using LSTM

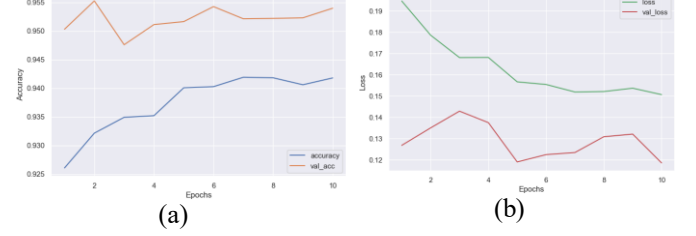


Fig. 8. (a) Accuracy. (b) Loss of multiclass classification using BiLSTM

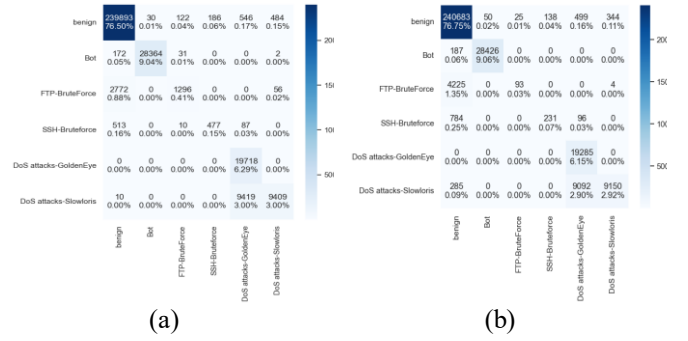


Fig. 9. (a) LSTM. (b) BiLSTM confusion matrix for multiclass classification

TABLE VII. COMPARISON OF THE EXISTENCE DL APPROACH WITH THE CSE-CIC-IDS2018 DATASET

Model	Accuracy	Label Type
LSTM [10]	93.87%	Binary
RNN[10]	93.85%	
GRU[10]	93.87%	
RC-NN[24]	94.28%	Multiclass
Our BiLSTM	98.62%	
Our LSTM	98.30%	
LSTM[12]	94%	
CNN[25]	91.5%	
Our BiLSTM	95.40%	
Our LSTM	95.01%	

E. Discussion And Comparison With Other Results

As mentioned above our proposed models LSTM and BiLSTM were tested using CSE-CICIDS2018 Datasets in both binary and multiclass classification. In general overview of the results, our models perform better in binary than multiclass classification, both LSTM and BiLSTM have achieved accuracy above 98% on binary, however in multiclass classification, our models obtain an accuracy of around 95.5%. On the other hand,

the outcome demonstrates that the BiLSTM model increases the LSTM model's accuracy by 0.32% and 0.39%, respectively, on binary and multiclass classification. Also, BiLSTM predictions on such labels are more accurate than LSTM, and Fig.10 confirms our analysis. Finally, from Table VII, it can be seen that our proposed LSTM and BiLSTM models performed better than previously studied DL approaches for IDS in the same dataset.

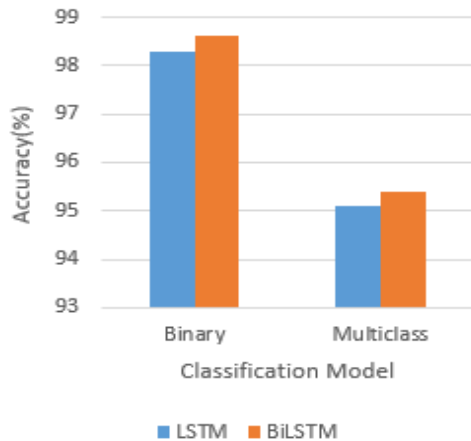


Fig. 10. Accuracy comparison of BiLSTM and LSTM on both Binary and Multiclass classification

V. CONCLUSION AND FUTURE WORK

This study attempts to develop a Deep Learning-based powerful IDS to detect intruders in IoT networks. The proposed model relies on two variants of RNN: LSTM and BiLSTM models for binary and multiclass classification using the CSE-CICIDS2018 dataset, a recent comprehensive network traffic dataset. Following the trial, the BiLSTM model performed better than the LSTM model in terms of accuracy, recall, precision and F1 score. The proposed multiclass and binary classification models outperformed earlier DL implementations for IDS. In the future, we look forward to further research on other Deep Learning methods for IDS in IoT networks, employing various optimization strategies to improve the detection capabilities of these models for large amounts of data. We also plan to improve prediction accuracy by running our experiments on real IoT applications using more efficient GPUs.

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